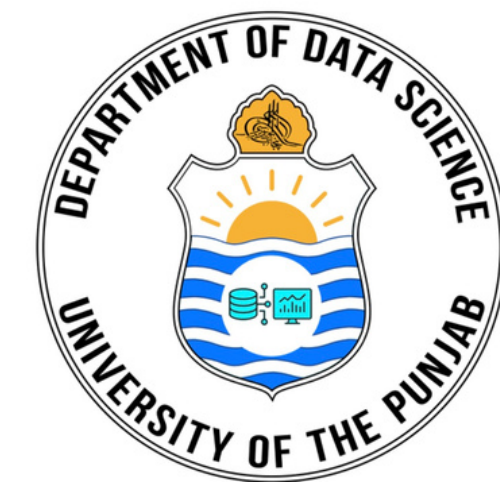


WEB SCRAPER BOT

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ABSTRACT

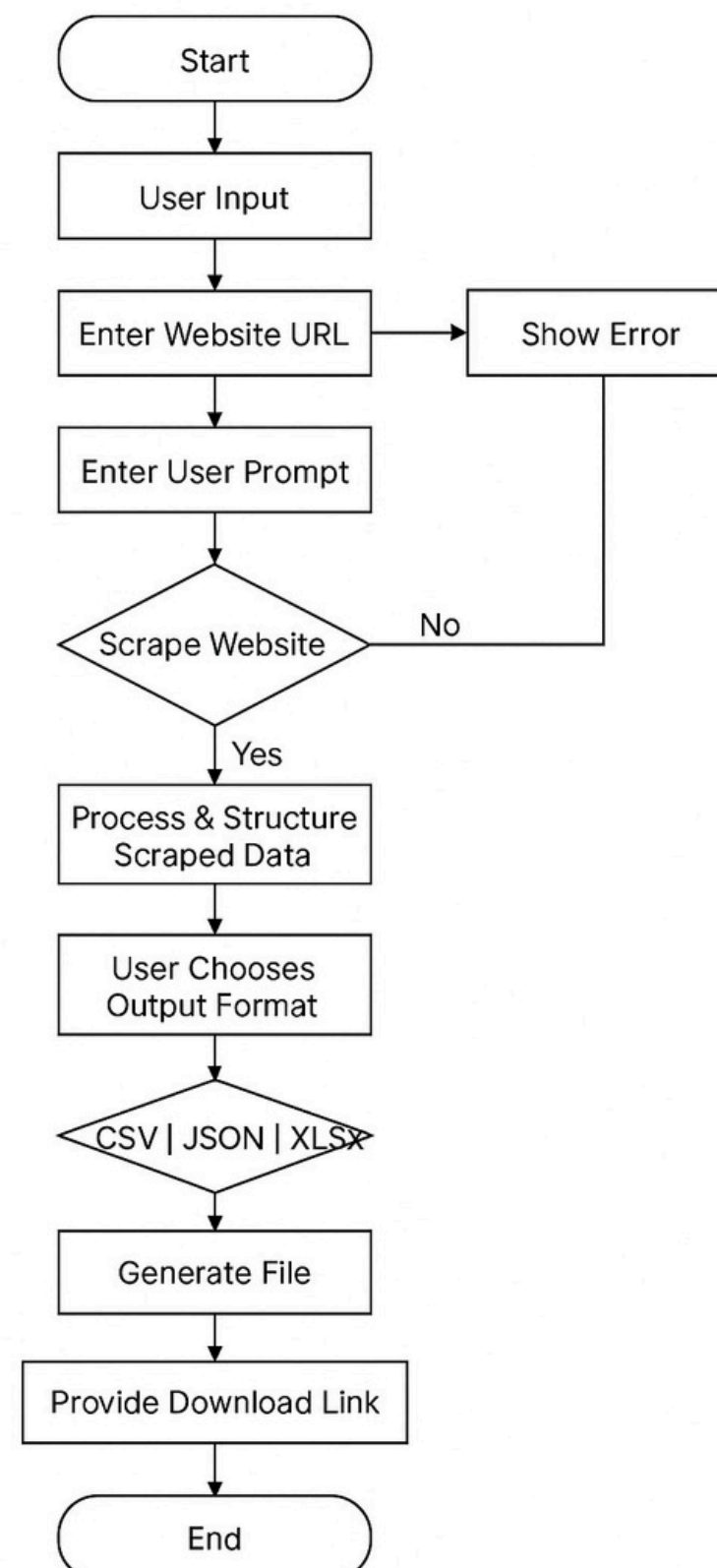
This project develops a Python-based chatbot that takes a URL and user prompts to scrape data from websites, then formats it into CSV, JSON, or XLSX for local download. It combines web scraping, data processing, and chatbot design to provide a simple, time-saving tool for non-technical users.

OBJECTIVES

- Design a chatbot interface for user interaction.
- Implement web scraping using Python libraries (BeautifulSoup, Selenium, Scrapy).
- Convert scraped data into structured formats (CSV, JSON, XLSX).
- Handle user errors like invalid URLs interactively.
- Provide easy-to-access downloadable data links.

METHODOLOGY

The project uses Python with BeautifulSoup, Selenium, and Scrapy for web scraping, spaCy for NLP, and Pandas with OpenPyXL for data export. A web-based interface ties everything together for easy, interactive data extraction and download.



RESULTS

- Successful extraction of data from multiple websites
- Data exported in selected formats (CSV, JSON, XLSX)
- Bot intelligently responded to invalid input
- Enhanced accessibility for non-programmers.
- Limitations with some CAPTCHA-protected and highly dynamic websites.

scrape this url <https://www.lennoxpro.com/hvac/furnaces/gas-furnaces/c/109> and give output in json
Apr 25 6:10 PM

Scraping completed successfully! Found 209 items.

Here's a summary of the scraped data:

```
1. { "Product Page url": "https://www.lennoxpro.com/el297uh045xe368-45btuh3t/p/24X60", "Product Image url": "https://images.lennoxpro.com/is/image/lennoxintl/24X60?product_related$1", "Image_Large": "https://images.lennoxpro.com/is/image/lennoxintl/24X60", "Product Name": "Lennox EL297UH045XE368 Furnace", "Product Title": "Lennox Elite EL297UHE, EL297UH045XE368, 97% AFUE, 44000 BTU, 3 Ton, 2 Stage, Constant Torque Upflow/Horizontal Low Nox Gas Furnace", "Product Listing Title": "Lennox EL297UH045XE368 2 Stage Furnace", "Brand": "Lennox", "Catalog Number": "24X60", "Model/Part Number": "EL297UH045XE368", "Base Unit of Measure": "Unit(S)", "Family": "EL297UHE", "Series": "Elite", "Heating Capacity": "44000 Btu/h", "Heating Efficiency": "97% AFUE", "Configuration": "Upflow/Horizontal", "Blower Size": "3 Ton", "Gas Stages": "2 Stage", "Blower Type": "Constant Torque", "Product Voltage": "115 VAC 1 Ph 60 Hz", "Product Depth": "27.75 inch", "Product Width": "17.5 inch", "Product Height": "33 inch", "Gross Weight": "129 lbs.", "Type": "2-Stage Constant Torque Horizontal Gas Furnace", "Energy Star Certified": "Yes", "Low Nox": "Low Nox" }
2. { "Product Page url": "https://www.lennoxpro.com/el297uh110xv60c110btuh5ton/p/24X54", "Product Image url": "https://images.lennoxpro.com/is/image/lennoxintl/24X54?product_related$1", "Image_Large": "https://images.lennoxpro.com/is/image/lennoxintl/24X54", "Product Name": "Lennox EL297UH110XV60C Furnace", "Product Title": "Lennox Elite EL297UHV, EL297UH110XV60C, 97% AFUE, 110000 BTU, 5 Ton, 2 Stage, Variable Speed Upflow/Horizontal Gas Furnace", "Product Listing Title": "Lennox EL297UH110XV60C 2 Stage Furnace", "Brand": "Lennox", "Catalog Number": "24X54", "Model/Part Number": "EL297UH110XV60C", "Base Unit of Measure": "Unit(S)", "Family": "EL297UHV", "Series": "Elite", "Heating Capacity": "110000 Btu/h", "Heating Efficiency": "97% AFUE", "Configuration": "Upflow/Horizontal", "Blower Size": "5 Ton", "Gas Stages": "2 Stage", "Blower Type": "Variable Speed", "Product Voltage": "115 VAC 1 Ph 60 Hz", "Product Depth": "27.75 inch", "Product Width": "17.5 inch", "Product Height": "33 inch", "Gross Weight": "137 lbs.", "Type": "2-Stage Variable Speed Upflow/Horizontal Gas Furnace" }
```

2.Table: Tool Comparison (Web Scraper Bot vs Manual vs Other Tools)

Feature	Web Scraper Bot	Manual Scraping	Octoparse
Requires Coding	✗	✓	✗
Free to Use	✓	✓	✗
Ease of Use	✓ (Very Easy)	✗ (Hard)	✓ (Easy)
Dynamic Site Support	Partial	✓	Partial

REFERENCES

<https://docs.scrapy.org/en/latest/>
<https://www.crummy.com/software/BeautifulSoup/bs4/doc/>
<https://www.selenium.dev/documentation/>